# 12.17 Banach's Matchboxes

WORKSHEET

**The question**

**STEP II 1992, Question 16**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q16

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

**Ask yourself** For the driver to discover on the $N$th reach that the toffees have run out, what must the $N$th reach be? What must the previous $N-1$ reaches contain?

### STAGE 1 Discovering the toffees are gone

**Ask yourself** In what order can the first $N-1$ reaches occur? What condition on $N-1-n$ ensures she never found the mint packet empty?

### STAGE 3 The conditional probability

**Ask yourself** We are told that a discovery happens on the $N$th reach. What is the probability that it was a toffee?

**Pause and try** Check the formula when $n = m$ and $p = q = \frac{1}{2}$. By symmetry the answer should be $\frac{1}{2}$ for every possible $N$. Is it?

## What's really going on

### Banach's matchboxes

In Banach's version, each box starts with $n$ matches and each pocket is equally likely. When he first finds a box empty, let $K$ be the number of matches in the other box. For $K = k$, he must have taken $n$ from the empty box and $n - k$ from the other, in any order, before reaching for the empty box once more; and either box could be the empty one:

$$\mathrm{P}(K = k) = 2\binom{2n-k}{n}\left(\frac{1}{2}\right)^{2n-k+1} = \binom{2n-k}{n}\left(\frac{1}{2}\right)^{2n-k}.$$

These probabilities decrease with $k$, except that $\mathrm{P}(K = 0) = \mathrm{P}(K = 1)$ (Figure 12.17.1). The expected number left is about $2\sqrt{n/\pi} - 1$: with 50 matches in each box, about 7. The $\sqrt{n}$ is the typical size of the imbalance between the two boxes after $2n$ random choices, the same square-root law as in the birthday problem (Section 9.9) and the random walk.

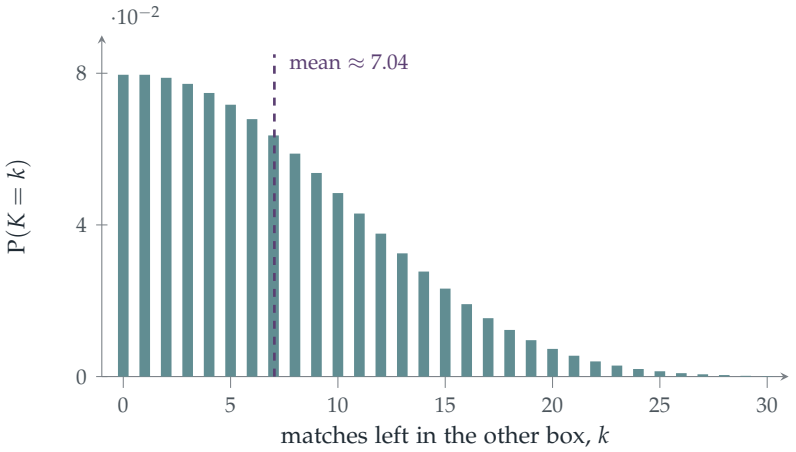



**Figure 12.17.1** Banach's matchbox problem with 50 matches in each box: the distribution of the number left in the other box when one is first found empty. The most likely values are 0 and 1, equally, but the mean is about 7.

### The negative binomial distribution

The number of trials needed to obtain the $r$th success in independent trials, each succeeding with probability $p$, has the *negative binomial distribution*: $\mathrm{P}(N = k) = \binom{k-1}{r-1} p^r q^{k-r}$. The taxi driver's problem is a race between two negative binomial processes (the toffees and the mints), each needing one more "success" than it has sweets. The negative binomial appears in insurance (numbers of claims), ecology (clustered counts of animals) and genetics, often as a more flexible alternative to the Poisson distribution.

#### Banach and the Scottish Café

Stefan Banach, one of the founders of functional analysis, did much of his mathematics in the Scottish Café in Lwów (now Lviv), where he and his colleagues wrote problems in a notebook kept by the waiter, the *Scottish Book*, with prizes ranging from a beer to a live goose. Hugo Steinhaus, Banach's mentor, devised the matchbox problem for a speech honouring Banach's smoking. The problem was first analysed in print by William Feller in his celebrated probability textbook of 1950.

## Try this next

1. **Small case.** With $n = m = 1$ and $p = \frac{1}{2}$, list the possible sequences and find the distribution of $N$.
2. **Banach's formula.** Verify that $\sum_{k=0}^{n} \binom{2n-k}{n} 2^{-(2n-k)} = 1$ for $n = 1$ and $n = 2$.
3. **Equal tops.** Show that $\mathrm{P}(K = 0) = \mathrm{P}(K = 1)$ for every $n$ in Banach's problem.
4. **Which runs out first?** For the taxi driver, find the probability that the toffees are the first packet to be discovered empty (at any time), when $n = m = 2$ and $p = \frac{2}{3}$.

## Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter VI, which introduced Banach's problem to the world.
- **R. Daniel Mauldin (ed.), *The Scottish Book*** (Birkhäuser). The problems from the Scottish Café, with commentary.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on the negative binomial distribution.

The full guided solution is in Section 12.17:
https://beyondstep.keithmcnulty.org/ch12/matchboxes.html